



CSA0911 – Programming in JAVA

CyberSentry AI: Intelligent Threat Investigation and Detection

Guided By

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Problem Statement

- **Rise in Cyber Threats:** Increasing phishing attacks, scam messages, malicious URLs, and social engineering attacks pose serious security risks.
- **Inefficient Traditional Detection:** Existing security systems rely on predefined rules and often fail to detect new or sophisticated cyber threats.
- **Lack of Explainable Analysis:** Most tools detect threats but do not explain why the content is malicious, reducing user trust and understanding.
- **Time-Consuming Manual Investigation:** Security analysts spend significant time manually analyzing suspicious emails, messages, URLs, and files.
- **Poor Risk Assessment and Response:** Users struggle to determine the severity of threats and take appropriate actions such as verifying, reporting, or blocking them.



Project Scope



- **Multi-Source Threat Analysis** - Supports the investigation of suspicious SMS, emails, URLs, text, and images from multiple sources.
- **AI-Powered Threat Detection** - Identifies phishing, scams, malicious URLs, and social engineering attacks using AI and Machine Learning.
- **Explainable Risk Assessment** - Generates risk scores, confidence levels, and evidence to explain why a threat is classified as suspicious.
- **Centralized Security Management** - Provides a unified dashboard for threat monitoring, investigation history, reports, alerts, and recommended actions.



Objectives



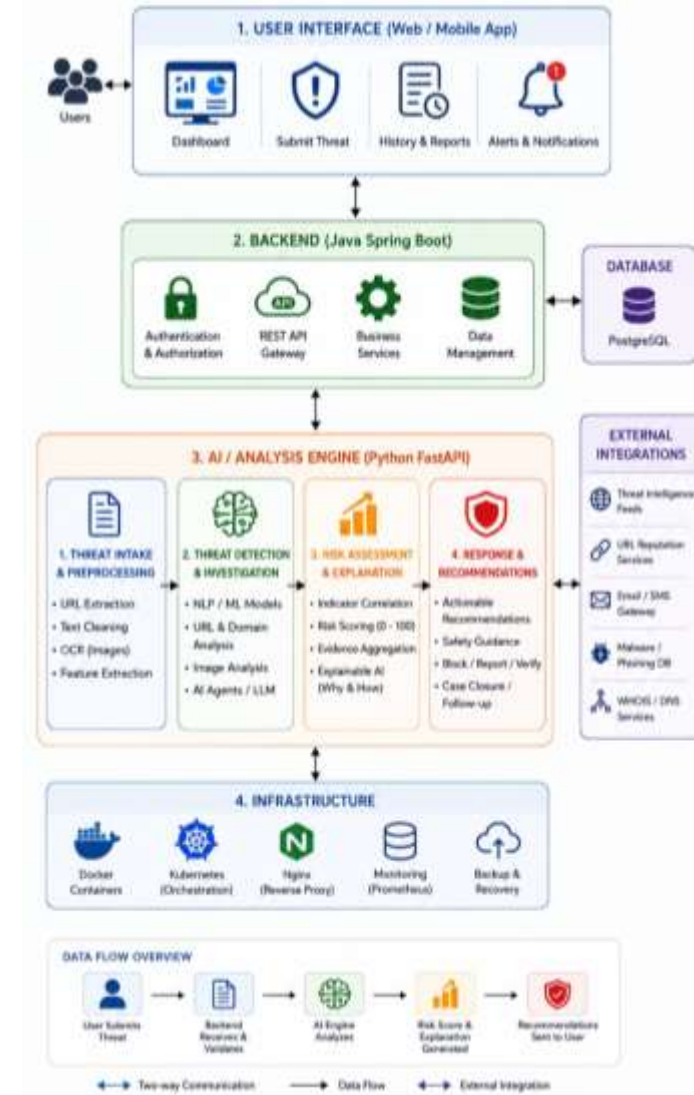
- Develop an Intelligent Threat Investigation System - Build an AI-based platform to detect and analyze cyber threats accurately.
- Automate Threat Detection and Risk Analysis - Use AI, NLP, and Machine Learning to classify threats and assign risk scores automatically.
- Provide Explainable Security Insights - Generate transparent reports with supporting evidence, confidence scores, and threat severity.
- Enhance User Security and Decision-Making - Deliver actionable recommendations such as Ignore, Verify, Report, or Block through an interactive security dashboard.

Literature Review / Technical Background

S. No.	Existing Work / Technology	Description	Limitation	Improvement in CyberSentry AI
1	Traditional Antivirus Systems	Detect threats using signature-based methods.	Cannot detect new or zero-day attacks effectively.	Uses AI to detect both known and unknown threats.
2	Machine Learning for Threat Detection	Applies ML algorithms to identify phishing and malicious content.	Limited explainability of predictions.	Provides explainable AI with evidence and confidence scores.
3	URL & Email Analysis Systems	Analyze URLs and emails to identify phishing attacks.	Focus only on specific input types.	Supports SMS, emails, URLs, text, and images.
4	Threat Intelligence Platforms	Collect and correlate threat information from multiple sources.	Complex to use and requires expert knowledge.	Presents simplified results through an interactive dashboard.
5	Explainable AI (XAI) in Cybersecurity	Explains AI decisions for better trust and transparency.	Limited integration with complete threat investigation workflows.	Combines AI detection, risk scoring, evidence generation, and user recommendations in one system.

METHODOLOGY / SYSTEM ARCHITECTURE

- **Collect Threat Data**
Receive SMS, emails, URLs, text, and images.
- **Preprocess Input**
Clean, validate, and extract useful data.
- **AI Threat Analysis**
Detect phishing, scams, and malicious content.
- **Classify Threats**
Identify the type of cyber threat.
- **Calculate Risk Score**
Determine the threat severity.
- **Generate Explanation**
Provide evidence and confidence score.
- **Display Results**
Show reports on the security dashboard.
- **Recommend Actions**
Suggest Ignore, Verify, Report, or Block.



Evaluation Metrics

- **Detection Accuracy** – Percentage of correctly identified threats.
- **Precision** – Measures how many detected threats are actually malicious.
- **Recall** – Measures how many actual threats are successfully detected.
- **F1-Score** – Balances precision and recall for overall performance.
- **Risk Score Accuracy** – Correctness of AI-generated risk assessment.
- **Response Time** – Time taken to analyze and classify a threat.
- **False Positive Rate** – Percentage of safe content incorrectly flagged as malicious.
- **User Satisfaction** – Effectiveness of recommendations and dashboard usability.

Expected Outcomes

- **Accurate Threat Detection** – Identifies phishing, scams, and malicious content with high accuracy.
- **Automated Threat Investigation** – Reduces manual effort through AI-powered analysis.
- **Risk Score Generation** – Assigns a reliable risk score and threat severity.
- **Explainable AI Reports** – Provides clear reasons and evidence for each detection.
- **Real-Time Security Insights** – Delivers fast analysis and investigation results.
- **Actionable Recommendations** – Suggests actions such as Ignore, Verify, Report, or Block.
- **Centralized Security Dashboard** – Displays alerts, reports, and investigation history in one place.
- **Improved Cybersecurity Awareness** – Enables users to make informed and secure decisions against cyber threats.



CONCLUSION



- CyberSentry AI helps detect cyber threats quickly.
- It uses AI to identify phishing and scam attacks.
- The system provides accurate risk analysis.
- It explains why a threat is detected.
- The dashboard shows clear reports and alerts.
- It helps users take the right security actions.
- It improves cybersecurity awareness and protection.

The background of the image is an abstract composition of flowing, wavy, and layered textures. The colors range from deep, dark browns and blacks to lighter, golden-brown and cream tones, creating a sense of depth and movement. The textures resemble liquid, smoke, or perhaps the grain of wood in a dynamic, swirling pattern. The overall effect is organic and fluid.

Thank you